Sparse production owner harmonic coercivity and compact-annulus boundary

TECT verification-first repository
claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-08-04 · this version issued 2026-08-04 · v1.0

1. Purpose and scope

R-164 reduces the direct T-050 analytic input to the complete-owner estimate

$$K_{\text{owner}} \succeq -\left(\frac{10}{11} - \rho\right) G_{\text{src}}, \qquad \rho > 0. \tag{1.1}$$

R-153 supplies the first actual strict-past conditional production fibre on which (1.1) can be tested. This note takes that production calculation one step further without replacing it by another abstract owner lemma.

On the fixed side-16 torus retain one antipodal past pair at p, one at $2p$, and one fresh final pair at $4p$, where $p = 2\pi/16$ points along one coordinate axis. Let

$$g = (g_p, g_{2p}) \in \mathbb{R}^{24}, \qquad G = |g|, \qquad \alpha \in \mathbb{R}^{12} \tag{1.2}$$

be whitened source coordinates. For the exact R-153 endpoint-plus-sixth conditional Hessian, excluding the explicit source Hessian, this note proves

$$\boxed{K_{\text{owner}}(g)[\alpha, \alpha] \geq P(G)|\alpha|^2,} \tag{1.3}$$

where

$$\begin{aligned} P(G) = {}& \frac{3125}{126241210368} G^4 - \frac{81135}{43679744} G^2 \\ & - \frac{8001}{8192000} G - \frac{177075}{174718976}. \end{aligned} \tag{1.4}$$

Exact rational derivative certificates then give

$$\boxed{K_{\text{owner}}(g) \succeq -\frac{9}{10}\mathrm{I} \quad \text{if} \quad 0 \leq G \leq 21 \quad \text{or} \quad G \geq 274.} \tag{1.5}$$

Since $10/11 - 1/110 = 9/10$, (1.5) is exactly the R-164 target with $\rho = 1/110$ on these two amplitude regions. Thus the noncompact amplitude question for this fixed twelve-dimensional fibre is reduced to the bounded open annulus

$$21 < G < 274. \tag{1.6}$$

This is not a proof on the annulus, not a complete multi-root production owner, and not T-050 closure. It is fixed-cutoff, fixed-floor, exactly nonaliased, and specific to the displayed (p:2p:4p) support. No BCC or other morphology is assumed or selected. No phase or PDE verdict occurs.

2. Exact conditional fibre

Condition on the (p,2p) strict past and write

$$W = m + X, \qquad n_i = \partial_i m, \qquad z = S_{4p}\alpha, \qquad u_i = \partial_i z, \tag{2.1}$$

where (X) is the centered fresh (4p) field. R-153 proves that the exact owner diagonal is

$$\begin{aligned} K_{\mathrm{owner}}(g)[\alpha,\alpha] = \int \sum_i \Big[& u_i^T \overline{B} u_i + 2u_i^T \overline{DB[z]} n_i \\ & + \frac{1}{2} \operatorname{Tr}\Big(\overline{D^2 B[z,z]} (n_i n_i^T - \Gamma_{<,i}) \Big) \Big] \,\mathrm{d}x \\ + \frac{9}{10} \int & \mathbb{E}\big[|W|^4 |z|^2 + 4|W|^2 (W \cdot z)^2 \big] \,\mathrm{d}x. \end{aligned} \tag{2.2}$$

The future-current covariance has already cancelled against its A7 trace owner. The strict-past covariance remains with the minus sign shown. The positive $u^T B u$ term and the second sixth-power term will be dropped once. The (9/10) source Hessian is not part of (2.2) and is not counted here.

R-130 gives the floor-uniform pointwise derivative envelopes

$$|D(B:Q)[a]| \le L_6 \|Q\|_{\mathrm{op}} |w| |a|, \qquad \left| \frac{1}{2} D^2 (B:Q)[a,a] \right| \le H_6 \|Q\|_{\mathrm{op}} |a|^2, \tag{2.3}$$

with

$$L_6 = \frac{1143}{250 P_X} < \frac{1143}{1000}, \qquad H_6 = \frac{7083}{500 P_X} < \frac{7083}{2000}, \qquad P_X = 4 + 10^{-12}. \tag{2.4}$$

The half-Hessian factor is already contained in H_6.

3. Sharp harmonic coercivity

Let $\theta = p \cdot x$. The real six-component fields have Fourier supports

$$\operatorname{supp} \widehat{m} \subset \{\pm 1, \pm 2\}, \qquad \operatorname{supp} \widehat{z} \subset \{\pm 4\}. \tag{3.1}$$

For normalized phase average $\langle \cdot \rangle$, put $r = |m|^2$, $f = r^2$, and $q = |z|^2$.

Lemma 3.1 (sharp (5/6) separation)

For every pair satisfying (3.1),

$$\boxed{\langle |m|^4 |z|^2 \rangle \ge \frac{5}{6} \langle |m|^4 \rangle \langle |z|^2 \rangle.} \tag{3.2}$$

The constant is sharp.

Proof. The top coefficient of r is r_4. Reality and Cauchy–Schwarz on the $2p$ Fourier coefficient give $r_0 \ge 2|r_4|$. Since $f_8 = r_4^2$, Parseval gives

$$f_0 \ge r_0^2 + 2|r_4|^2 \ge 6|f_8|. \tag{3.3}$$

The function q has only frequencies $0, \pm 8$, and positivity gives $q_0 \ge 2|q_8|$. Therefore

$$\langle fq \rangle = f_0 q_0 + 2\Re(f_8 \overline{q_8}) \ge f_0 q_0 - 2|f_8||q_8| \ge \frac{5}{6} f_0 q_0. \tag{3.4}$$

For scalar $m = \cos(2\theta)$ and $z = \sin(4\theta)$, the three averages are $3/8, 1/2, 5/32$, so equality holds. $\square$

For centered Gaussian (X), direct moment expansion gives

$$\mathbb{E}|m + X|^4 = |m|^4 + 2|m|^2 \operatorname{Tr} C + 4m^T C m + (\operatorname{Tr} C)^2 + 2\operatorname{Tr}(C^2) \ge |m|^4. \tag{3.5}$$

Combining (3.2), (3.5), and spatial Jensen supplies the positive G^4 term in (1.4).

4. Exact spectral enclosures

The internal three-complex-component mass matrix is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \tag{4.1}$$

Exact principal minors give

$$\frac{7}{250}\mathrm{I} < M \le \frac{27}{100}\mathrm{I}. \tag{4.2}$$

With $s(x) = x^2 + Zx + r$, $Z = -0.9252754126$, and $r = 0.4740336473$, the covariance is

$$\Gamma_q = \operatorname{diag}\left((s(|q|^2)\mathrm{I} + M)^{-1}, (s(|q|^2)\mathrm{I} + M)^{-1}\right). \tag{4.3}$$

Using the classical $333/106 < \pi < 355/113$, monotonicity of s on the three relevant rational brackets, and the registered global symbol floor gives

$$\frac{100}{63}\mathrm{I} \le \Gamma_p, \Gamma_{2p} \le \frac{100}{31}\mathrm{I}, \tag{4.4}$$

$$\frac{20}{91}\mathrm{I} \le \Gamma_{4p} \le \frac{10}{43}\mathrm{I}, \qquad \operatorname{Tr}\Gamma_{4p} \le \frac{60}{43}. \tag{4.5}$$

For example, the rational endpoint bounds for the largest past operators are

$$s((333/106)^2/64) + \frac{27}{100} = \frac{6313708938155690251}{10099815680000000000} < \frac{63}{100}, \tag{4.6}$$

$$s((355/113)^2/16) + \frac{27}{100} = \frac{7223409606693823849}{13043788880000000000} < \frac{14}{25}. \tag{4.7}$$

The corresponding (4p) endpoints lie respectively below $91/20$ and above $43/10$. Thus no floating eigenvalue enters the certificate.

5. Owner lower polynomial

Exact antipodal synthesis and Fourier orthogonality give

$$\int |m|^2 \,\mathrm{d}x \ge \frac{100}{63}G^2, \quad \int |m|^2 \,\mathrm{d}x \le \frac{100}{31}G^2, \quad \int |n|^2 \,\mathrm{d}x \le 4p^2\frac{100}{31}G^2, \tag{5.1}$$

and, for the final tangent,

$$|z|^2 \le \frac{2(10/43)}{V}|\alpha|^2, \quad |z||u| \le \frac{8|p|(10/43)}{V}|\alpha|^2, \quad \int |z|^2 \,\mathrm{d}x \ge \frac{20}{91}|\alpha|^2. \tag{5.2}$$

Here $V = 16^3 = 4096$. The fresh field contributes

$$\int |n|\mathbb{E}|X| \,\mathrm{d}x \le 2|p|\sqrt{2(100/31)(60/43)}G. \tag{5.3}$$

The strict-past current covariance obeys

$$\|\Gamma_<\|_{\mathrm{op}} \le \frac{10p^2(100/31)}{V} < \frac{625}{507904}. \tag{5.4}$$

Lemma 3.1 and (5.1)–(5.2) give the sextic coefficient

$$A = \frac{3}{4}\frac{(100/63)^2(20/91)}{4096^2} = \frac{3125}{126241210368}. \tag{5.5}$$

The two (DB) orientations, the current-mean half-Hessian, and the retained past trace give the safe adverse constants

$$\begin{aligned} B_2 &= \frac{p_*^2(100/31)(10/43)}{V}\left(32\frac{\cdot 1143}{1000} + 8\frac{\cdot 7083}{2000}\right) = \frac{81135}{43679744}, \\ B_1 &= \frac{32(1143/1000)p_*^2(10/43)(301/100)}{V} = \frac{8001}{8192000}, \\ D_0 &= \frac{7083}{2000}\frac{\cdot 10}{43}\frac{625}{507904} = \frac{177075}{174718976}. \end{aligned} \tag{5.6}$$

where $p^2 < p_*^2 = 5/32$ and $\sqrt{2(100/31)(60/43)} < 301/100$. The past covariance is spatially constant in this sharp common-even chart, so Parseval gives $\int |z|^2 \leq (10/43)|\alpha|^2$ in D_0; no extra antipodal factor is present, and H_6 already owns the half-Hessian. Substitution into (2.2) proves (1.3)–(1.4).

6. Compact-annulus certificate

On $0 \leq G \leq 21$,

$$P'(G) \leq 4A21^3 - B_1 = -\frac{25427}{425984000} < 0, \tag{6.1}$$

and

$$P(21) + \frac{9}{10} = \frac{145670815281}{2271346688000} > 0. \tag{6.2}$$

Thus (P(G)>-9/10) throughout the inner interval. At the outer endpoint,

$$P'(274) = \frac{1338311264185381}{1314683854848000} > 0, \qquad P''(274) = \frac{65160500885}{3505823612928} > 0, \tag{6.3}$$

and (P”) increases for positive (G). Hence (P’) stays positive for all $G \geq 274$. Finally,

$$P(274) + \frac{9}{10} = \frac{1847465221877063}{2629367709696000} > 0. \tag{6.4}$$

Equations (6.1)–(6.4) prove (1.5).

The unresolved set (1.6) is open, not compact. Its closure $21 \leq G \leq 274$, together with unit directions in $\mathbb{S}^{23} \times \mathbb{S}^{11}$, is compact; the two radial endpoints are already certified by (6.2) and (6.4). At the fixed positive floor its matrix entries are continuous. A rigorous interval, Bernstein, or rational-SOS subdivision of that closed certification domain can therefore decide this fibre without any amplitude tail assumption. A finite floating scan cannot replace that certificate.

7. Route decision and next falsifier

The useful advance is structural: the sixth-power stabilizer does control large realized past on this production fibre, and the exact source-plus-owner target also holds on a certified inner ball. The only unresolved amplitudes for this fibre are $21 < G < 274$. The next bounded computation should build the exact 12×12 R-153 matrix on the compact closed certification domain $21 \leq G \leq 274$ and use outward interval errors. A single certified eigenvalue below (-10/11) would refute the uniform R-164 target; a pass proves only this fibre and must not be lifted to all roots or nonlinear feedback without a separate uniform argument.

The exploratory floating quadrature used while choosing this route is not evidence: it sampled finitely many past directions, used non-certified quadrature, and found no value remotely near the adverse threshold. Its only role was to prioritize an analytic tail certificate rather than assert a positive result.

8. Devil's-advocate review

1. **Objection: the (5/6) factor is an unproved decorrelation assumption. DISMISSED.** Lemma 3.1 is an exact Fourier coefficient calculation, and its scalar equality fixture proves sharpness.
2. **Objection: an antipodal or six-real normalization factor was lost. DISMISSED.** Equations (5.1)–(5.4) use $S_q S_q^* = 2\Gamma_q/V$, the final derivative $4|p|$, both DB orientations, the fresh-field trace, and the uncancelled past covariance.
3. **Objection: H_6 needs another factor $1/2$. DISMISSED.** R-130 defines H_6 for the half-Hessian already; (2.3) retains that factor firewall.
4. **Objection: (1.5) closes the complete production owner. UPHELD against that inference.** It treats one fixed sparse conditional principal fibre. Cross-root blocks, other past harmonics, nonlinear/revisit controls, low variables, and uniform removal remain open.
5. **Objection: the unverified numerical scan is proof of the annulus. UPHELD against that inference.** Only an interval or exact compact certificate can promote the annulus.
6. **Objection: success or failure here selects BCC or a physical PDE. UPHELD against that inference.** This is a conditional mathematical fibre of one candidate action and carries no morphology or physical-vacuum conclusion.

9. Reproduction and Result footer

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Result ID: A13-CLASSII-SPARSE-PRODUCTION-OWNER-HARMONIC-
  COERCIVITY-COMPACT-ANNULUS-BOUNDARY (R-165).
Precise statement: on the fixed exact p:2p past / fresh-4p conditional fibre,
  K_owner[g] >= P(|g|) I with the rational polynomial (1.4), and
  K_owner >= -9I/10 for |g|<=21 or |g|>=274.
Dependencies: A1; R-130, R-153, R-164.
Scope: fixed side-16 finite chart, positive floor, exact continuum torus
  average, unit regulator multipliers at p, 2p, and 4p, only the declared
  p:2p strict-past and fresh-4p pair, and whitened antipodal sources.
Evidence grade: ANALYTIC, EXACT, EXECUTED, AUDITED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_sparse_production_owner_harmonic_
  coercivity_compact_annulus_boundary.py
Independent command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_sparse_production_owner_harmonic_
  coercivity_compact_annulus_boundary_independent.py
Integrated command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_sparse_production_owner_harmonic_
  coercivity_compact_annulus_boundary_verify.py
Expected output: primary 38/38, independent 30/30, integrated, PDF, and
  release checks all PASS.
Falsification gate: any failure of the 5/6 Fourier lemma, covariance
  enclosure, owner factor ledger, rational polynomial, derivative signs,
  authority hashes, PDF, record registration, or release verification.
Tier before / after: T4 / T4; no promotion.
Proof complete: false. T-050 closed: false. Sector A closed: false.
No-overclaim statement: no certification of the open annulus, complete production
  owner, feedback/removal, T-050, A13, phase, morphology, PDE, or Sector A.
Next required action: interval-certify [21,274] or find a counterdirection on 21<G<274,
  then test the nonzero complete multi-root owner separately.
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